

Research and IPM

Models: Diseases

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Crop: Grape

Disease: Powdery Mildew

Pathogen: *Uncinula necator*

Note: Before using a model that was not field tested or validated for a specific location, a model should be tested for one or more seasons under local conditions to verify that it will work in the desired location. See " Validation Work" below.

Model 1 of 2

Model developer and citation

Thomas, C. S., Gubler, W. D., and Leavitt, G. 1994. Field testing of a powdery mildew disease forecast model on grapes in California. *Phytopathology* 84:1070 (abstr.).

Weber E., Gubler, D., and Derr, A. Powdery mildew controlled with fewer fungicide applications. *Practical Winery & Vineyard*, January/February 1996.

Sensor location

In the fruit zone in the canopy.

Input variables

Environmental: Hourly average temperature, daily maximum temperature; hourly leaf wetness duration (ascospore model only).

Calculated: Daily average temperature; daily value of number of hours that $70 \leq T \leq 85$ F, amount of time that temperature is above 95 F.

Model description

This model contains two stages based on pathogen biology, an ascospore and a conidial stage. Check with your local Cooperative Extension agent to determine if both stages are important in your area.

Ascospore stage

To determine ascospore infection risk levels, the model calculates the daily average temperature and measures the hours of leaf wetness. The model modifies the Mills table developed for apple scab ascospore infection by predicting infection based on 2/3 of the hours of required wetness (column 2 in Table 1).

Table 1. Modified Mills Table: Hours of wetting required for infection for ascospore infection.

Daily Average Temperature (F)	Hours of Leaf Wetness required for heavy ascospore infection (2/3 original Mills value)	Leaf Wetness Hours required for heavy infection (original Mills value)



42	40	60
43	34	51
44	30	45
45	27.3	41
46	25.3	38
47	23.3	35
48-49	20	30
50	19.3	29
51	18	27
52	17.3	26
53	16.7	25
54-55	16	24
56-57	14.7	22
58-59	14	21
60-61	13.3	20
62	12.7	19
63-75	12	18
76	12.7	19
77	14	21
78	17.3	26

Conidial stage

1. The start of the conidial stage of the model is triggered early in the growing season by three consecutive days with six consecutive hours of temperatures between 70 and 85 F; for each of these three days, the model assigns 20 points to the disease risk index.
2. The conidial infection index increases by 20 points on each subsequent day where at least six continuous hours of temperatures between 70 and 85 F occur.
3. If there are less than six consecutive hours of temperatures between 70 and 85 F, 10 points are subtracted from the index. This is the case if the temperature falls below 70 or goes above 85 for more than 45 minutes, since this amount of time is considered to break the accumulation of conducive temperature hours.
4. If the temperature is 95 F or higher for at least 15 minutes, 10 points are subtracted from the index for the day.
5. If on the same day with 6 continuous hours between 70-85 F the temperature exceeds 95 F for 15 minutes or more, then the model adds 20 points for the six hours but subtracts 10 points due to the high temperature. Thus, for that day the model adds 10 points.
6. If after subtracting a day's points the index is less than zero, reset the index to zero. If after adding a day's points the index is greater than 100, reset the index to 100.
7. On any one day the index should not decline by more than 10 points or increase by more than 20 points.

Action threshold

According to the model, for **ascospore infection** treat when the conditions required for heavy infection have occurred (at the daily average temperature, treat when the number of modified Mills leaf wetness hours have occurred, column 2 of Table 1 (2/3 of the number of Mills' hours of leaf wetness for severe infection). For example, at a daily average temperature of 55 F, 16 hours of leaf wetness are required for heavy infection (2/3 of 24 hours of leaf wetness for severe infection from original Mills Table). At a daily average temperature of 65 F, 12 hours are required for heavy infection (2/3 of 18 hours of leaf wetness from original Mills Table).



For **conidial infections**, the timing of treatments is based on temperature, fungicide type, and spray intervals (see Table 2). An index of 30 or less indicates that a spray interval can be stretched to the label maximum. An index of 40 to 50 indicates that a spray interval can be of intermediate length. An index of 60 to 100 indicates that there is high pressure for powdery mildew and spray intervals should be shortened to the label minimum. After treatment, the index is reset to zero.

Table 2. Treatment timing guidelines based on risk index and spray material.

Powdery Mildew Risk Index	Spray Material	Spray Interval
0 to 30	sulfur dust	14 days**
	micronized sulfur	18 days**
	DMI fungicides*	21 days**
40 to 50	sulfur dust	10 days
	micronized sulfur	14 days
	DMI fungicides*	17 days
60 to 100	sulfur dust	7 days
	micronized sulfur	10 days
	DMI fungicides*	14 days
* Demethylation inhibitors such as Bayleton, Rally, Rubigan.		
** Or label maximum.		

Model validation

Since 1995 validation work has been conducted in multiple growing regions of California. In 1997, California PestCast is sponsoring validation projects in additional table and wine grape areas and in raisin grapes. The model is also being validated in New York, Washington, and Oregon; Germany, Austria, and Australia.

Model implementation

This model is being implemented by University of California Extension Plant Pathologist Doug Gubler, growers, Cooperative Extension advisors, and licensed consultants in several grape-growing counties in California.

Model 2 of 2

Model developer and citation

R. L. Snyder, La Vine, P. D., Sall, M. A., Wrysinski, J. E., and Schick, F. J. Grape Mildew Control in the Central Valley of California Using the Powdery Mildew Index. U. C. Cooperative Extension Leaflet 21342 (April, 1983).

Sensor location

At regional CIMIS weather stations.

Input variables

Environmental: measured or forecasted daily high and low temperatures, rain of greater than 0.10 inches.

Model description

This model was developed for wine and raisin grapes. It is based on the assumption that host tissue growth and weathering of sulfur are the two main reasons why sulfur must be reapplied in vineyards. Daily mildew indexes (DMI) are calculated from daily minimum and maximum temperatures (see Table 1). DMI are accumulated to determine the powdery mildew index (PMI). DMI begin accumulating twelve days after initial leaf appearance or 6-inch shoot growth (whichever comes first) and continue throughout the season until the berries contain 12-15% sugar.

Table 1. Daily mildew indexes for various high and low daily temperatures.

Daily low temperature	Daily high temperature --- F° 										
	55-60	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100	100-105	105-110

40-45	0.083	0.083	0.083	0.083	0.083	0.077	0.067	0.067	0.056	0.05 0	0.043
45-50	0.083	0.083	0.083	0.083	0.091	0.083	0.077	0.067	0.059	0.05 3	0.048
50-55	0.083	0.083	0.083	0.083	0.100	0.091	0.083	0.077	0.063	0.05 9	0.053
55-60	0.083	0.083	0.083	0.091	0.111	0.100	0.091	0.083	0.077	0.06 3	0.059
60-65	----	0.083	0.083	0.111	0.111	0.111	0.100	0.091	0.077	0.071	0.067
65-70	----	----	0.100	0.143	0.143	0.125	0.125	0.100	0.091	0.077	0.071
70-75	----	----	----	0.143	0.167	0.143	0.125	0.111	0.091	0.077	0.067
75-80	----	----	----	----	0.14 3	0.125	0.091	0.083	0.067	0.056	*
80-85	----	----	----	----	----	0.091	0.077	0.059	*	*	*
85-90	----	----	----	----	----	----	0.059	0.059	*	*	*

* Under high temperature conditions, the amount of sulfur applied per acre should be reduced to avoid excess leaf burn. If possible, sulfur should be applied in the evening after temperatures drop below 90° F to allow slow oxidation during the night, thus avoiding some leaf burn which occurs during the hotter daytime hours.

Table 2. Examples of how to use the Powdery Mildew Index (PMI)

June	25	6.72
	26	6.81
	27	6.92
	28	7.01
	29	7.13
	30	7.25
July	1	7.34
	2	7.42
	3	7.51
	4	7.60
	5	7.73*
	6	7.84*
	7	7.92*
	8	8.02*
*PMIs are forecasted for dates July 5-8 on July 5.		
Example 1: Last dusting - 6/25 $6.72+1=7.72$ Next dusting - 7/5	Example 2: Last dusting - 6/27 $6.92+1=7.92$ Next dusting - 7/7	Example 3: Last dusting - 6/29 $7.13+1=8.13$ Next dusting - ?

Action threshold

According to the model, initiate first dusting twelve days after initial leaf appearance or 6-inch shoot growth (whichever comes first).

Subsequent dustings should occur when the difference between the current PMI and the PMI on the last dusting date equals or exceeds 1.0 (see [Table 2](#) for an example). When precipitation exceeds 0.10 inch, the vineyard should be re-dusted. 

Model validation

Sall, M. A., Wrysinski, J. E., and Schick, F. J. 1983. Temperature-based timing method of sulfur dust applications. California Agriculture 37, #5 & 6.

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